



Technical report

CP020 · BBS Mexico · Aug 5, 2026

Operational record

Project

CP020 · BBS Mexico

Work performed by

Gabriel Santos

Worker email

gabriel.santos@example.test

Report created by

Gabriel Santos

Creator email

gabriel.santos@example.test

Client

IMPC Gmbh

Date

Aug 5, 2026

System

Main Assembly Cell PLC

Site

BBS Saltillo Plant

Area / line

Assembly Line 2

Station / machine

Station 10 - 14 Transfer

Status

Approved

Safety-related

Yes

Technical records

System type

PLC & Safety Controller

PLC platform

**Siemens S7-1500 & Allen
Bradley GuardLogix**

Controller

CPU 1517F-3 PN/DP

HMI / SCADA

**Siemens WinCC Unified &
FactoryTalk View**

Network protocol

**PROFINET / EtherNet/IP
Industrial Gateway**

Software version

**TIA Portal v19 / Studio
5000 v35**

Program reference

BBS_MEX_L2_MAIN_REV_3.4

Change summary

Problem / symptom

**Intermittent emergency
stop trip on conveyor
index transfer**

Diagnosis / root cause

**Dual-channel safety relay
pulse test timing
discrepancy between
safety gate and PLC
safety input module**

Change performed

**Adjusted discrepancy
time window in Safety
Evaluation configuration
from 50ms to 200ms per
manufacturer
recommendation**

Production impact

**Zero unplanned downtime
observed over 4 hours of
test production**

Validation

**Tested 10 consecutive
emergency stop triggers
and re-engaged safety
circuit; all fault codes
reset cleanly**

Validation result

**Pass · Fully functional and
within safety SIL3
compliance standards**

Open risk

None identified; hardware and firmware signatures backed up

Rollback plan

Restore safety project file BBS_MEX_L2_MAIN_REV_3.3 from local backup PLC SD card

Technical changes

DATE	DETAIL	CHANGE SUMMARY	STATUS
Aug 5, 2026	Safety Input Module F-DI 16x24VDC	Updated input filter parameter and discrepancy time window	Approved
Aug 5, 2026	HMI Alarm Banner	Added explicit diagnostic text: Station 12 Gate Discrepancy Fault	Approved